

Project Initialization and Planning Phase

Date	29 September 2026
Team ID	6ab2a427dbd693aadc5a7d4b
Project Name	Student Enrollment & Course Management System (SECMS)
Maximum Marks	2 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform loan approval using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced risks, and happier customers. Key features include a machine learning-based credit model and real-time decision-making.

Project Overview	
Objective	The primary objective is to digitalize and automate student enrollment & course management process, ensuring faster admission, zero overbooking, secure Razorpay payment, and transparent course allocation.
Scope	Covers full lifecycle: Online registration, course catalog with live seats, prerequisite check, real-time seat locking, fee payment, admin verification, faculty assignment, and reporting dashboard. Future: mobile app.

Problem Statement

Description	Current manual enrollment system causes long queues, manual form filling, data entry errors, seat overbooking, lost documents, payment not reflecting, no real-time seat status and heavy admin workload during admission season.
Impact	Solving these will result in 80% faster enrollment, zero data loss, improved student satisfaction, reduced admin workload, accurate reporting, and increased transparency in admission process

Proposed Solution

Approach	Employing web-based 3-tier architecture with React.js frontend, Node.js with Express.js backend, MySQL database and AWS cloud deployment to create dynamic, scalable and secure enrollment platform.
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Resource Requirements

Resource Type	Description	Specification / Allocation
Computing Resources	CPU / GPU specs, number of cores	Intel i5, 4 cores, No GPU needed
Memory	RAM specifications	8 GB min (16 GB server)
Storage	Disk space for data logs	500 GB SSD
Frameworks	JS frameworks	React.js, Bootstrap 5, Node.js + Express.js
Libraries	Additional libraries	axios, bcrypt, jwt, multer, mysql2, nodemailer, Razorpay SDK
Development Environment	IDE version control	VS Code, Chrome DevTools, Postman, GitHub
Data Source	Source size format	College Records, Course Catalog CSV/MySQL, 10000+ records
Cloud Hosting	Hosting	AWS EC2 t2.medium, S3 Bucket, RDS MySQL